Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

1. INTRODUCTION

1.1 Project Overview

This project focuses on automating the process of importing employee data from an external system into ServiceNow. The system enables seamless integration of employee details such as Name, Email, Department, Manager, and Phone Number. Additionally, it enhances the Incident Management process by automatically displaying relevant employee-related information using dot walking.

1.2 Purpose

The purpose of this project is to:

- Reduce manual data entry
- Ensure accurate and up-to-date employee records
- Improve incident handling efficiency
- Enable real-time data access using reference fields

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2. IDEATION PHASE

2.1 Problem Statement

Organizations often manage employee data in external systems. Manually updating this data in ServiceNow leads to:

- Data inconsistency
- Increased workload
- Errors in employee information
- Delay in incident resolution

2.2 Empathy Map Canvas

Users: HR Team, IT Support Team

- **Think & Feel:** Want accurate and quick access to employee details
- **See:** Multiple disconnected systems
- **Say & Do:** Manually update records, verify employee info

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- **Pain:** Time-consuming, error-prone
- **Gain:** Automation, real-time data, improved efficiency

2.3 Brainstorming

Possible solutions:

- Automated import using CSV/XLSX files
- Integration APIs
- Scheduled data sync
- Use of Transform Maps and Import Sets

Final Idea Chosen:
Use ServiceNow Import Sets and Transform Maps to automate employee data import.

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. HR prepares employee data in Google Sheets
2. File is downloaded as CSV/XLSX
3. File uploaded into ServiceNow
4. Data transformed into sys_user table
5. IT uses updated data in Incident Management

3.2 Solution Requirement

- Employee data file (CSV/XLSX)
- ServiceNow instance
- Import Set table (import_employee)
- Transform Map
- Incident form customization

3.3 Data Flow Diagram

External File → Import Set Table → Transform Map → sys_user Table → Incident Form

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3.4 Technology Stack

- Google Sheets (Data preparation)
- ServiceNow (Platform)
- CSV/XLSX (Data format)

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4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution addresses:

- Manual data entry issues
- Data inconsistency
- Lack of automation

4.2 Proposed Solution

- Create employee dataset
- Import into ServiceNow
- Map fields using Transform Map
- Use dot walking in Incident form

4.3 Solution Architecture

External Data → Import Set → Transform Map → User Table → Incident Module

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Task	Duration
Data Preparation	1 Day
Import Set Creation	1 Day

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Task	Duration
Transform Map Setup	1 Day
Testing	1 Day
Documentation	1 Day

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Verify import speed
- Check data accuracy
- Validate reference field mapping
- Ensure Incident form loads quickly

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7. RESULTS

7.1 Output Screenshots (You will add)

- Google Sheet Data
- Import Set Table
- Transform Map Configuration
- sys_user Table Records
- Incident Form with auto-filled details

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual work
- Improves data accuracy
- Faster incident resolution

- Real-time data visibility

Disadvantages

- Initial setup required
- Dependency on correct data format
- Needs periodic updates

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9. CONCLUSION

The project successfully automates employee data import into ServiceNow and enhances Incident Management using dot walking. It improves efficiency, reduces errors, and ensures real-time access to employee information.

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10. FUTURE SCOPE

- API-based real-time integration
- Scheduled automatic imports
- Advanced reporting dashboards
- Integration with HR systems

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11. APPENDIX

Source Code (if any)

(Not required for this project)

GitHub & Project Demo Link

https://github.com/ashwinj032-jpg/Automating-Data-Import-and-Relationship-Mapping-Using-Import-Sets-Dot-Walking

https://drive.google.com/file/d/1yO7gLasK9oFONrf67EtS2E3MwsaLoZb2/view?usp=sharing